

The Dimension Barrier: Separating q -SDH from Constant-Size Assumptions up to Bounded Native Freshness

A typed catalogue and audited implication graph, the generic-representation boundary, and a fixed-representation meta-reduction

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¹Independent research collective, math.st · July 2026

ABSTRACT

A q -type assumption such as q -SDH hands the adversary a power ladder $g, g^x, \dots, g^{x^q}$ whose length grows with a scheme’s usage parameter. Problem P2.2 asks for one of two outcomes: a black-box reduction from q -SDH to a **constant-size** assumption, or a meta-reduction proving that no such reduction exists. We deliver the four required artifacts and locate the honest boundary between them. First, a uniform typed catalogue of the inversion, exponent, and gap families, together with an audited implication graph in which every edge carries its breaker transformation, call count, and loss. Second, an exact account of where a direct q -SDH-to-SXDH reduction dies: a constant-size decisional challenge exposes only a fixed-dimensional exponent span, and already the prefix $([1]_1, [x]_1, [x^2]_1)$ cannot be embedded. Third, an audit of the Lu–Zhandry separation: no fully black-box generic-representation reduction bases prime-order q -SDH on a true fixed-size assumption once $n < q - 1$ (one group) or $\binom{n+2}{2} < q$ (bilinear), and we prove that any standard-oracle reduction whose guarantee is uniform over all group representations is already inside that class. Fourth, our sharpest new theorem: for one named concrete representation, a fully black-box reduction that introduces at most s_1 unexplained (“fresh”) labels in the q -SDH source group is impossible under hardness of the fixed assumption whenever $n_1 + s_1 < q - 1$; a safe overcounted threshold for broader target-valued bilinear q -type games is $\binom{n+s+2}{2} + t < q$. Three attempted extensions fail for identifiable reasons, and we record their post-mortems: random relabeling breaks at a representation quantifier, efficiently seeded sparse encodings are information-theoretically distinguishable from random injections, and structured-label density does not bound execution-wise freshness. The unrestricted problem — reductions with native-label rank linear in q , and reductions that read the adversary’s code — remains open, and we state it as such.

Keywords. q -SDH · q -type assumptions · meta-reductions · black-box separations · generic group model · pairing-based cryptography

1 Introduction

Pairing-based cryptography leans, in large parts of its deployed and standardized corpus, on assumption **families** parametrized by an integer q that grows with usage. The prototype is the strong Diffie–Hellman assumption of Boneh and Boyen [1]: in a group of prime order r , given the power ladder $g, g^x, g^{x^2}, \dots, g^{x^q}$, it must be infeasible to output any pair $(c, g^{1/(x+c)})$. The parameter is not cosmetic. In the Boneh–Boyen short-signature proof, q equals the adversary’s signing-query budget [1]; in the BBS family it tracks the number of issued signatures, and recent concrete-security work shows the dependence is real in both directions [2], [3]; and Cheon’s algorithms show that the assumption **quantitatively degrades** as the ladder lengthens, by roughly $\sqrt{d}$ bits of security for favorable divisors d of $r \pm 1$ [4]. A q -type assumption is therefore an awkward foundation: a family whose strength deteriorates in exactly the parameter that deployments increase.

Problem P2.2 poses the natural structural question. **Either** exhibit a black-box reduction from q -SDH to a constant-size assumption — one whose instance size does not grow with q — **or** prove via meta-reduction that no such reduction exists. The required deliverables are a uniform-notation catalogue, an implication graph with cited edges and loss factors, a written account of where an attempted constant-size reduction breaks, and a precisely scoped meta-reduction statement, proved or stated as a target.

This paper is the record of three research sessions on that problem, and its finding is a **scoped impossibility** — strictly more than the published boundary, strictly less than an unrestricted separation.

Main finding (scoped). The strongest justified deliverable is a scoped impossibility, not a reduction and not an all-black-box impossibility. **CITED** On the published side, no fully black-box generic-representation reduction bases prime-order q -SDH on a true fixed-size assumption whose challenger emits n group elements, once $n < q - 1$ in a single group or $\binom{n+2}{2} < q$ with a bilinear map [5]. **PROVED** Our sharpest new theorem (Theorem 16, from attempt A006) extends the separation to **one named concrete representation**: a fully black-box reduction that introduces at most s_1 unexplained native labels in the q -SDH source group is impossible under hardness of the fixed assumption whenever $n_1 + s_1 < q - 1$; for broader target-valued bilinear q -type games a safe overcounted threshold is $\binom{n+s+2}{2} + t < q$. **PROVED** What remains open is exactly the residual class: reductions whose native-label rank grows linearly with q , and reductions that use the q -SDH adversary’s code non-black-box.

1.1 Contributions and honest scope

We contribute (i) the typed catalogue and audited implication graph with per-edge losses (§3–4, Figure 1); (ii) the catalogue of constant-size prime-order candidates and what each does **not** supply (§5); (iii) Cheon’s quantitative obstruction restated in typed notation (§6); (iv) the exact break point of the direct SXDH embedding — an exponent-span argument that kills the attempt already at x^2 (§7); (v) an audit of the Lu–Zhandry generic-representation separation and a new boundary corollary: every **representation-uniform** fully black-box standard-oracle reduction is already a generic-representation reduction and inherits the published thresholds (§8); (vi) the bounded-native-freshness meta-reduction, our main new theorem, with its bilinear sharpening and safe overcount (§9, Figure 2, Figure 3); (vii) post-mortems of three dead extensions — random relabeling,

efficient sparse encodings, and density-to-freshness transfer — each with its exact failure point (§10); and (viii) a final scope classification separating what is ruled out from what is open (§11–12).

We state the scope plainly, as the problem’s ground rules demand. Nothing here reduces q -SDH to a constant-size prime-order assumption, and nothing here is an unrestricted black-box impossibility. The status of P2.2 is **partial**: complete for the catalogue, the published generic boundary, and the bounded-freshness representation-dependent class; open for linear native rank and for non-black-box use of the adversary.

2 Setting and typed notation

CITED Throughout, r is prime, $e : \mathbb{G}_1 \times \mathbb{G}_2 \rightarrow \mathbb{G}_T$ is a non-degenerate bilinear map on groups of order r , and g_1, g_2 are fixed generators. We write $[z]_1 = g_1^z$, $[z]_2 = g_2^z$, and $[z]_T = e(g_1, g_2)^z$ for $z \in \mathbb{Z}_r$ [1].

Definition 1 (typed power ladder). **PROVED** For $I \subset \mathbb{Z}_{\geq 0}$, the typed power ladder is

$$P_i(x; I) = ([x^j]_i)_{j \in I}, \quad (1)$$

where $[x^0]_i = [1]_i$ is the named base. “Search” means output the named value; “decision” means distinguish the named value from an independent uniform element of the same typed group. The two tasks are never silently interchanged.

The typing discipline matters more than it may appear: the names DHE, strong DH, BDHE, and their index shifts are not canonical across the literature, so every claim below refers to a displayed tuple, never to a bare acronym [4], [5]. Near-name variants are not identified without tuple equality.

3 The assumption catalogue

The first deliverable is a uniform-notation catalogue (sub-goal SG-01, complete). Table 1 lists the inversion and strong-DH family; Table 2 lists the exponent and gap family. Each row records the exact instance and target in typed notation, with the epistemic tag under which the research log holds it.

Table 1: The inversion and strong-DH family in typed notation (SG-01).

Name	Exact instance and target
CITED q -SDH (source core)	Input $P_1(x; \{0, \dots, q\})$; output any $(c, [1/(x+c)]_1)$ with $c \in \mathbb{Z}_r$, $x+c \neq 0$ [1], [5].
CITED q -SDH (typed Boneh–Boyen)	The source core plus $[1]_2, [x]_2$; the extra $\mathbb{G}_2$ terms permit public verification but do not alter the search relation [1].
CITED q -DHI	Input $P_1(x; \{0, \dots, q\})$; output $[1/x]_1$ — the prescribed choice $c = 0$ of the q -SDH relation [1].
PROVED q -DDHI	Input $P_1(x; \{0, \dots, q\})$ and $T \in \mathbb{G}_1$; decide whether $T = [1/x]_1$ or $T \leftarrow \mathbb{G}_1$ uniform.
CITED q -BDHI	Symmetric original: input $(g, g^x, \dots, g^{x^q})$, output $e(g, g)^{1/x}$. Typed lift: input $P_1(x; \{0, \dots, q\})$ and $[1]_2$; output $[1/x]_T$ [6], [7].
PROVED q -DBDHI	Decision version of the typed target: distinguish $T = [1/x]_T$ from uniform $T \leftarrow \mathbb{G}_T$.

Name	Exact instance and target
CITED q -wBDHI	Input independent generators g_1, h_2 and $P_1(x; \{1, \dots, q\})$; output $e(g_1, h_2)^{1/x}$ [7].
CITED q -wBDHI*	Same input as q -wBDHI; output $e(g_1, h_2)^{x^{q+1}}$ [7].

Table 2: The exponent and gap family in typed notation (SG-01). The gap tuple omits the power x^q that its target exponentiates.

Name	Exact instance and target
CITED q -DHE	Input $P_1(x; \{0, \dots, q\})$; output $[x^{q+1}]_1$ [5].
CITED q -DDHE	Input $P_1(x; \{0, \dots, q\})$ and $T \in \mathbb{G}_1$; decide whether $T = [x^{q+1}]_1$ or uniform. Also called q -strong DDH in some sources [5].
CITED q -BDHE (gap form)	Input independent g_1, h_2 and $[x^j]_1$ for $j \in \{0, \dots, q-1, q+1, \dots, 2q\}$; output $e(g_1, h_2)^{x^q}$ [4], [7].
PROVED q -DBDHE	Decision version of the preceding gap target in $\mathbb{G}_T$.
CITED q -aBDH	Input $P_1(x; \{0, \dots, q\})$, $[1]_2, [x]_2$, an independent h_2 , and $h_2^{x^{q+2}}$; output $e(g_1, h_2)^{x^{q+1}}$ [1].

4 The implication graph and its loss audit

PROVED An arrow $H(A) \rightarrow H(B)$ means “hardness of A implies hardness of B ”: operationally, a breaker for B is transformed into a breaker for A . Every edge in Figure 1 was audited against the primary reduction text — not a survey summary — and carries its transformation, call count, and loss in Table 3.

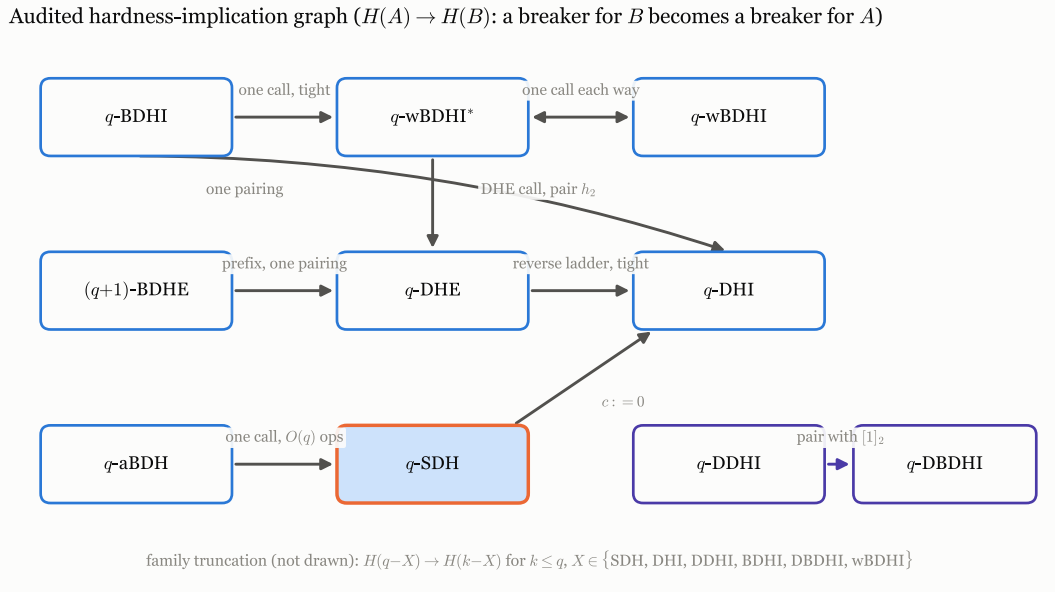


Figure 1: The audited implication graph (SG-02), drawn from the edge table below; schematic rendering of proved/cited reductions, not data. Solid computational family in blue; the decisional pair at lower right; q -SDH highlighted as the problem’s anchor. Every edge is one oracle call and success-preserving. The truncation family $H(q-X) \rightarrow H(k-X)$, $k \leq q$, applies to $X \in \{\text{SDH, DHI, DDHI, BDHI, DBDHI, wBDHI}\}$ but not to targets whose exponent depends on q (DHE, wBDHI*, gap-BDHE).

Table 3: Audited edge table (SG-02). ε denotes the invoked breaker’s success probability or advantage.

Edge	Breaker transformation	Loss
CITED $H(q\text{-aBDH}) \rightarrow H(q\text{-SDH})$	Run the q -SDH breaker to get $(c, [1/(x+c)]_1)$. Divide $X^{q+2} - (-c)^{q+2}$ by $X + c$ to get $X^{q+1} + w(X)$ with $\deg w \leq q$; pairing and the supplied $h_2^{x^{q+2}}$ isolate the wanted x^{q+1} term [1].	1 call; ε preserved; $O(q)$ ops
PROVED $H(q\text{-SDH}) \rightarrow H(q\text{-DHI})$	Feed the same ladder to the DHI breaker; return its output as the SDH answer with $c = 0$ [1].	1 call; ε ; $O(1)$
PROVED $H(q\text{-BDHI}) \rightarrow H(q\text{-DHI})$	Pair the DHI output $[1/x]_1$ with $[1]_2$ [1], [6].	1 call; ε ; 1 pairing
CITED $H(q\text{-BDHI}) \rightarrow H(q\text{-wBDHI}^*)$	From $(w, w^\beta, \dots, w^{\beta^q})$, give the wBDHI* breaker base w^{β^q} , the reversed ladder, and $h = w^s$; raise its answer to $1/s$ [7].	1 call; ε ; $O(q)$ ops
CITED $H(q\text{-wBDHI}^*) \leftrightarrow H(q\text{-wBDHI})$	Reverse the ladder with new base $g_1^{x^q}$ and hidden exponent $1/x$; the two targets transform into one another [7].	1 call each way; ε ; $O(q)$
PROVED $H(q\text{-wBDHI}^*) \rightarrow H(q\text{-DHE})$	Run the DHE breaker on $(g_1, g_1^x, \dots, g_1^{x^q})$ and pair its output $g_1^{x^{q+1}}$ with h_2 [4], [7].	1 call; ε ; 1 pairing
PROVED $H(q\text{-DHE}) \rightarrow H(q\text{-DHI})$	To use a DHI breaker, set base $g_1^{x^q}$, hidden exponent $1/x$, and reverse the ladder; its output is $g_1^{x^{q+1}}$ [4].	1 call; ε ; $O(q)$ reversal
PROVED $H((q+1)\text{-BDHE}) \rightarrow H(q\text{-DHE})$	On the gap tuple, run the q -DHE breaker on the prefix through x^q and pair $g_1^{x^{q+1}}$ with h_2 ; ignore higher powers [4].	1 call; ε ; 1 pairing
PROVED $H(q\text{-DDHI}) \rightarrow H(q\text{-DBDHI})$	Pair every source challenge with $[1]_2$ and invoke the DBDHI distinguisher [6].	1 call; advantage preserved; 1 pairing
PROVED $H(q\text{-X}) \rightarrow H(k\text{-X}), k \leq q$	Delete the unused suffix of the ladder before calling the k -breaker; valid for $X \in \{\text{SDH, DHI, DDHI, BDHI, DBDHI, wBDHI}\}$ [1].	1 call; preserved; $O(q)$ handling

PROVED The truncation edge formalizes why the assumptions strengthen as q grows: hardness at the larger parameter implies hardness at the smaller, while the converse is not supplied by tuple truncation. **CITED** Boneh and Boyen further prove q -SDH random self-reducible by rescaling the hidden exponent and bases — one solver call, success preserved, $O(q)$ exponentiations [1].

5 Constant-size prime-order candidates

Which fixed-size assumptions could serve as a target? Table 4 catalogues the standard candidates (SG-03), separating source-group, cross-source, and target-group consequences.

Table 4: Constant-size prime-order candidates (SG-03). For every true fixed-size candidate here, the generic separation of §8 rules out the fully black-box generic reduction from q -SDH once q crosses the dimension threshold; this is **not** a claim that any of these assumptions is false [5].

Assumption	Fixed-size typed experiment	What it supplies — and does not
CITED DDH in $\mathbb{G}_i$	Distinguish $([1]_i, [a]_i, [b]_i, [ab]_i)$ from $([1]_i, [a]_i, [b]_i, [z]_i)$.	A decisional degree-two correlation in one source group; no source power ladder [5].
CITED XDH / SXDH	XDH: DDH holds in the designated source group despite the pairing; SXDH: DDH in both $\mathbb{G}_1$ and $\mathbb{G}_2$.	Constant-size decisional assumptions for asymmetric pairings; no map from $\mathbb{G}_T$ back to a source group [8], [9].
CITED DLIN in $\mathbb{G}_i$	For independent u, v, h : distinguish $(u, v, h, u^a, v^b, h^{a+b})$ from the tuple with random last element.	A constant-size linear-relation challenge; still a fixed-dimensional source exponent span [10].
CITED co-CDH	Given $([1]_1, [a]_1, [1]_2)$, output $[a]_2$.	A constant-size cross-source search problem; pairing reveals $[a]_T$ but not $[a]_2$ in the typed interface [9], [11].
CITED DBDH	Given $([1]_1, [a]_1, [b]_1, [1]_2, [c]_2, T)$, distinguish $T = [abc]_T$ from uniform.	A constant-size target-group decision problem; it does not manufacture $[x^j]_1$ for unboundedly many j [6].

6 The quantitative obstruction: Cheon's exponent recovery

Even before any separation, the ladder itself leaks. The following two statements are Cheon's, restated in typed notation.

Proposition 2 (Cheon, $d \mid r - 1$). **CITED** If a divisor $d \leq q$ of $r - 1$ is available, the ladder contains g^x and g^{x^d} , and x is recoverable in $O(\log r (\sqrt{(r-1)/d} + \sqrt{d}))$ group operations [4].

Proposition 3 (Cheon, $d \mid r + 1$). **CITED** If $d \mid r + 1$ and the ladder reaches degree $2d$, then x is recoverable in $O(\log r (\sqrt{(r+1)/d} + d))$ group operations [4].

Corollary 4. **PROVED** Recovering x immediately solves q -SDH: choose any $c \neq -x$ and compute $[1/(x+c)]_1$ by known-scalar exponentiation.

CITED These attacks reduce generic square-root security by roughly a factor $\sqrt{d}$ for favorable divisors, so concrete group sizes must account for both q and the factorization of $r \pm 1$ [4]. This is the quantitative face of the same phenomenon the rest of the paper treats structurally: the ladder is a high-dimensional correlated object, and nothing constant-size manufactures it.

7 Where a direct reduction dies: the exponent-span barrier

The problem asks for a written account of where an attempted constant-size-assumption reduction breaks. Attempt A001 tried the most natural route: use one source-group DDH challenge — hence one component of an SXDH challenge — to synthesize the ladder a q -SDH adversary expects, and convert its answer into a DDH decision. The reduction class tested is straight-line, type-preserving, and algebraic: it combines challenge elements by group operations and known-scalar exponentiation, invokes one q -SDH adversary, and has no source-group encoding of a target-group pairing value.

Proposition 5 (exponent-span barrier). **PROVED** Write the source DDH challenge exponents as $(1, a, b, z)$, with $z = ab$ in the real branch and independent z in the random branch. Every $\mathbb{G}_1$

element such a reduction creates before its oracle call has exponent in $L = \text{span}_{\mathbb{Z}_r}\{1, a, b, z\}$. In the random branch, no nonconstant x satisfies both $x \in L$ and $x^2 \in L$; hence even the prefix $([1]_1, [x]_1, [x^2]_1)$ of a nontrivial q -SDH instance cannot be embedded.

Proof. In the random branch, treat a, b, z as algebraically independent. Put $x = A + Ba + Cb + Dz \in L$. If $x^2 \in L$, comparing the coefficients of $a^2, b^2, z^2, ab, az, bz$ — which do not occur in L — forces, over an odd prime field, $B = C = D = 0$. So x is a known constant, statistically independent of the challenge bit, and an oracle query built from it cannot transfer any distinguishing advantage. Pairing does not repair the embedding: it produces degree-two exponents in $\mathbb{G}_T$ only, while q -SDH needs the missing powers in $\mathbb{G}_1$, and the typed interface has no operation $\mathbb{G}_T \rightarrow \mathbb{G}_1$. $\square$

Remark 6. PROVED This kills the stated straight-line algebraic class at x^2 — before the first oracle call — but is not by itself a proof against rewinding, adaptive multiple calls, or representation-dependent and non-algebraic reductions. Those classes are the subject of the next two sections.

The failure is dimensional, and it is the germ of everything that follows: a constant-size challenge exposes a fixed-dimensional algebraic span, while the q -SDH ladder demands $q + 1$ correlated powers.

8 The generic-representation boundary

8.1 The Lu–Zhandry separation, audited

The fixed-dimensional-span failure of A001 was turned, by Lu and Zhandry, into a separation for a substantially broader class [5]. Attempt A002 audited the primary proof bodies; we restate the result in the repository’s terms. Methodologically the result descends from Coron’s meta-reduction technique for signature-scheme optimality bounds [12], which is ancestry, not itself a q -SDH separation.

Definition 7 (GR-BB reduction). CITED A generic-representation black-box (GR-BB) reduction is polynomial-time, fully black-box in a possibly **inefficient** adversary whose generic group-operation queries it may observe and answer, and representation-independent: it receives concrete bit-string labels and may apply arbitrary bit gates to them, but its advantage is defined as the infimum over all — possibly inefficient — implementations of the group. The formulation permits interactive assumptions and adaptive oracle calls [5].

Definition 8 (fixed-size and q -type assumptions). CITED A **fixed-size** assumption may be interactive and may contain non-group auxiliary data, but across the experiment its challenger outputs at most n group elements. A **q -type** assumption requires that a transcript expose at least q linearly independent bounded-degree polynomial functions tied to a hidden x_0 , such that knowing x_0 lets an efficient algorithm break the assumption with noticeable probability [5].

Theorem 9 (Lu–Zhandry). CITED In a prime-order generic group, if a GR-BB reduction maps such a q -type assumption to a fixed-size assumption of size $n < q - 1$, then a generic polynomial-time algorithm breaks the fixed-size assumption. In the generic bilinear extension the same conclusion holds when $\binom{n+2}{2} < q$ [5].

Proof. (Audited skeleton; the proof is Lu–Zhandry’s.) The meta-reduction selects an inefficient perfect q -adversary that brute-forces the hidden exponents. It traces every reduction-created source element as a known linear combination of the n fixed challenge elements, so the exponent vectors submitted to the adversary lie in a subspace of dimension at most $n + 1$. A Vandermonde rank argument and finite-field root finding leave only polynomially many candidates for x_0 ; generic equality tests identify the valid ones, so the otherwise inefficient adversary is simulated efficiently, and the reduction with its oracle replaced by this simulation is a generic attack on the fixed-size assumption. With a pairing, recorded exponents may be quadratic in the n input exponents; linearizing in all degree-at-most-two monomials gives dimension $\binom{n+2}{2}$ and the bilinear threshold [5]. $\square$

Corollary 10 (q -SDH instantiation). CITED q -SDH meets the q -type definition: its ladder supplies the required independent polynomials, and knowledge of x yields a valid answer by selecting any $c \neq -x$ after a consistency check. Hence no generic reduction from q -SDH to a true fixed-size assumption exists at the stated thresholds [5].

Remark 11 (why composite order escapes). CITED The prime-order proof step fails in unknown-composite-order groups: finding the required polynomial roots can be as hard as factoring, and independently rerandomizable hidden subgroups supply the “shadow” dimensions that the positive Déjà Q reductions exploit — computational q -SDH and broad polynomial q -type classes **do** reduce to constant-size subgroup hiding in composite-order pairing groups, with $\Theta(q)$ hybrid loss in the original framework and logarithmic loss in the follow-up [5], [13], [14]. The positive and negative results are consistent; the barrier is specifically prime-order.

CITED The theorem’s tracing argument also covers algebraic reductions whose outputs carry algebraic explanations, but it does not rule out every black-box reduction in every formalization: representation-specific standard-model reductions, and reductions non-black-box in the q -adversary, are outside its stated class [5]. Section 3.5 of the paper likewise leaves open derived groups whose law depends on auxiliary bit strings, equality branches, or concrete encodings [5]. These gaps are exactly where the remainder of this paper operates.

8.2 Representation-uniform reductions are already generic (A003)

One might hope that a **standard-oracle** reduction — a single machine that may inspect encodings as bit strings — escapes the GR-BB class. It does not, as long as its guarantee is uniform over representations.

Definition 12 (UR-FBB reduction). PROVED For single-stage prime-order group games F and Q , a reduction is **universally representation-uniform fully black-box** (UR-FBB) if a single PPT oracle machine R (i) treats the Q -adversary as an oracle, though it may run multiple copies and rewind; (ii) uses the group through labeling, group-operation, equality, and optional pairing interfaces, and otherwise performs arbitrary bit computation on labels; and (iii) satisfies pointwise representation uniformity: for **every** admissible implementation G of the group, including possibly inefficient ones, and every possibly inefficient adversary A , if $\text{Adv}_{Q^G}(A) \geq \varepsilon$ for non-negligible ε , then $\text{Adv}_{F^G}(R^{A,G}) \geq \delta$ for some non-negligible δ independent of G and A . Conditions (i)–(ii) match the classical fully-black-box interface [15].

Theorem 13 (boundary theorem). **PROVED** Every UR-FBB reduction is a GR-BB reduction. Consequently no UR-FBB reduction bases prime-order q -SDH on a true fixed-size assumption of size n when $n < q - 1$, nor, with a generic bilinear map, when $\binom{n+2}{2} < q$.

Proof. A GR adversary with non-negligible infimum advantage breaks Q^G for every implementation G with a common non-negligible lower bound ε . Applying uniformity pointwise gives one non-negligible δ with

$$\inf_G \text{Adv}_{F^G}(R^{A,G}) \geq \delta. \quad (2)$$

Conditions (i)–(ii) make R a GR oracle circuit, so this is a GR-BB reduction — indeed stronger than required, since the source definition may assign a reduction per adversary. The thresholds then follow from Theorem 9 and its q -SDH instantiation. For single-stage games the statement transfers to the type-safe model, since TS-BB and GR-BB reductions exist if and only if one another [5]. Label inspection does not invalidate the wrapper: for label length $\ell > \log_2 r + \omega(\log \lambda)$, a polynomially bounded interaction guesses a valid unseen label only with negligible probability [5]. $\square$

Remark 14. **PROVED** The qualifier **universal** is load-bearing: it includes possibly inefficient random sparse implementations. A guarantee promised only over efficiently computable or standardized representations need not include them and is **not** promoted to GR-BB by this argument. That weaker quantifier is precisely the residual case attacked next.

9 Main theorem: bounded native freshness

The remaining gap after §8 is the reduction promised for **one named concrete representation** G_* — an elliptic-curve group with its standardized encoding, say — where the machine may exploit native operations: validation, decompression, hash-to-group, table lookups, arbitrary encoding-bit branches. Random relabeling cannot reach this class (§10.1). Attempt A006 reaches part of it directly, without relabeling, by charging a new resource.

Definition 15 (s -fresh reduction). **PROVED** Fix a concrete prime-order group implementation G_* with efficient validity, equality, group operations, and known-scalar exponentiation, and a fixed-size assumption F^{G_*} whose challenger returns at most n source-group elements per execution. A PPT oracle reduction R is **s -fresh** if it is fully black-box in every, possibly inefficient, q -SDH adversary and, on every execution, at most s distinct valid source-group labels first appear without one of these explanations: challenger output, the public generator, known-scalar labeling, or a previously observed group operation. “First appear” is tested when a bit string is first used as a typed group element, including when it is supplied directly inside a q -SDH oracle query. Each such unexplained label consumes one **freshness unit**.

PROVED The budget is semantic and execution-wise; it need not be derivable from the source code of R . Native validation, decompression, hash-to-group, and arbitrary bit-dependent code are all permitted — the meta-reduction merely maintains a counter and invokes the theorem under the promise that the counter never exceeds s .

Theorem 16 (bounded-fresh-label separation). **PROVED** For polynomially bounded q, n, s with $n + s < q - 1$, an s -fresh PPT fully-black-box reduction from q -SDH to F^{G_*} yields a PPT

attack on F^{G^*} with the reduction's success probability against a perfect q -SDH adversary. Consequently, if F^{G^*} is hard, no such reduction exists at that threshold.

Proof. **Enlarged coefficient trace.** The meta-reduction runs R on the real F^{G^*} challenge and keeps a dictionary mapping every typed source label seen so far to a vector $D(P) \in \mathbb{Z}_r^{n+s+1}$, with coordinates for the at most n challenger labels, the at most s native fresh labels, and the public generator. A newly returned challenger label and a newly encountered fresh label each receive the next unused unit vector; known-scalar labeling receives the corresponding multiple of the generator vector; a group-operation output receives the sum or difference of its operands' vectors. If an output label already occurs in the dictionary, its **old** vector is retained.

Evaluation invariant. Let $b \in \mathbb{Z}_r^{n+s+1}$ hold the actual discrete logarithms of the basis labels relative to the public generator. Inductively every dictionary vector satisfies

$$\log_g P = \langle D(P), b \rangle. \quad (3)$$

Retaining the old vector after an actual-label collision preserves this: both candidate vectors evaluate to the same group element, so both evaluate to the same logarithm — the meta-reduction never needs to **discover** the hidden linear relation among the entries of b that the collision witnesses. Adaptivity and representation-bit branches do not disturb the invariant: the same concrete code runs on the same labels, coefficients may depend on all previously observed bits and equalities, and every unexplained valid label created along the chosen branch is charged to the freshness budget instead.

Simulating the inefficient adversary. In each oracle call, the labels of the proposed q -SDH ladder form the rows of a known matrix M with at most $n + s + 1$ columns. Let w be the actual discrete logarithm of the queried generator. For a valid nonidentity tuple with hidden exponent x , the invariant gives

$$M(b/w)^\top = (s_1(x), \dots, s_q(x))^\top, \quad (4)$$

so the actual x lies on the root list produced by the low-dimensional subspace root-list argument of Lu–Zhandry with $n + s$ replacing n [5] — only the **dimension of the column space** is used, never the logarithms of the fresh basis elements. When $n + s < q - 1$ the list is polynomial. Fix the perfect adversary to reject invalid or identity-generator tuples and use deterministic tie-breaking. For each candidate x the simulator checks the ladder equations by ordinary scalar exponentiation and equality in the real group; the nonidentity first ladder element makes a consistent x unique; the simulator picks a public $c \neq -x$ and returns the queried generator raised to $(x + c)^{-1}$. Rewinding and multiple adaptive calls are covered because the dictionary and freshness counter persist across the whole execution. The simulation is transcript-exact for the selected perfect adversary and adds only polynomial field arithmetic, root finding, and group operations, so it introduces no success-probability loss. $\square$

Figure 2 shows the architecture. The only imported mathematical ingredient is the root-list simulation and its bilinear quadratic lift; the theorem changes the column count, not the proof [5].

A006 meta-reduction: one coordinate per native fresh label restores the trace invariant

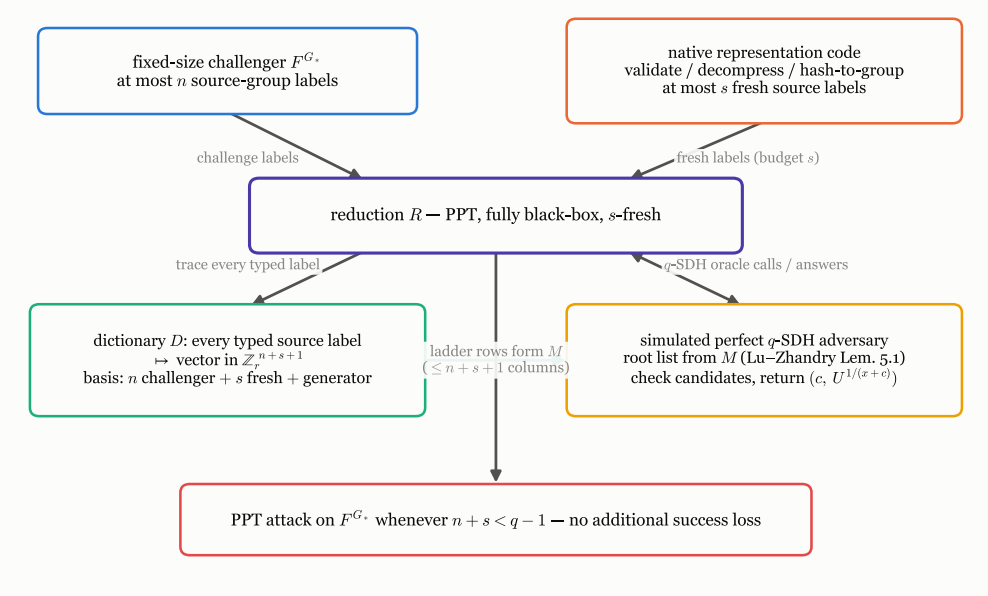


Figure 2: Architecture of the bounded-fresh-label meta-reduction (A006); structural schematic of the proof, not data. Native code may inject up to s unexplained source labels; each receives one formal coordinate, so every ladder row sent to the simulated adversary lies in a known $(n + s + 1)$ -dimensional span and the root-list simulation applies whenever $n + s < q - 1$.

9.1 Bilinear sharpening and safe overcount

Proposition 17 (source-valued pairing case). **PROVED** For typed q -SDH whose oracle instance lies in one source group, pairing outputs do not enlarge that source-group trace. If n_1 counts the fixed-assumption challenge labels that can reach that source through recorded source-to-source maps and s_1 counts unexplained labels in that source, the proof of Theorem 16 gives the sharper threshold

$$n_1 + s_1 < q - 1, \quad (5)$$

provided no unrecorded target-to-source conversion is available — if one is, each of its outputs is fresh.

Proposition 18 (broader bilinear games). **PROVED** For a bilinear q -type game whose independent polynomial family may occur in the target group, or as a conservative common-space bound, at most s unexplained source labels and t unexplained target labels give trace dimension at most

$$\binom{n + s + 2}{2} + t, \quad (6)$$

the quadratic term containing all degree-at-most-two monomials in the $n + s + 1$ source coordinates and each unexplained target label one additional coordinate. Replaying the bilinear root-list theorem yields the separation whenever $\binom{n+s+2}{2} + t < q$. The bound deliberately overcounts monomials that typing may forbid: it is a safe sufficient threshold, not a claimed tight one.

Corollary 19 (what a surviving reduction must do). **PROVED** Contrapositively, any fully-black-box fixed-representation reduction that coexists with hardness of F^{G_*} must, in some execution, introduce at least $q - 1 - n_1$ unexplained labels in the q -SDH source group, or use an unrecorded conversion into that source, or leave the fully-black-box class altogether. Linear-in- q native freshness and non-black-box access to the adversary remain outside the theorem.

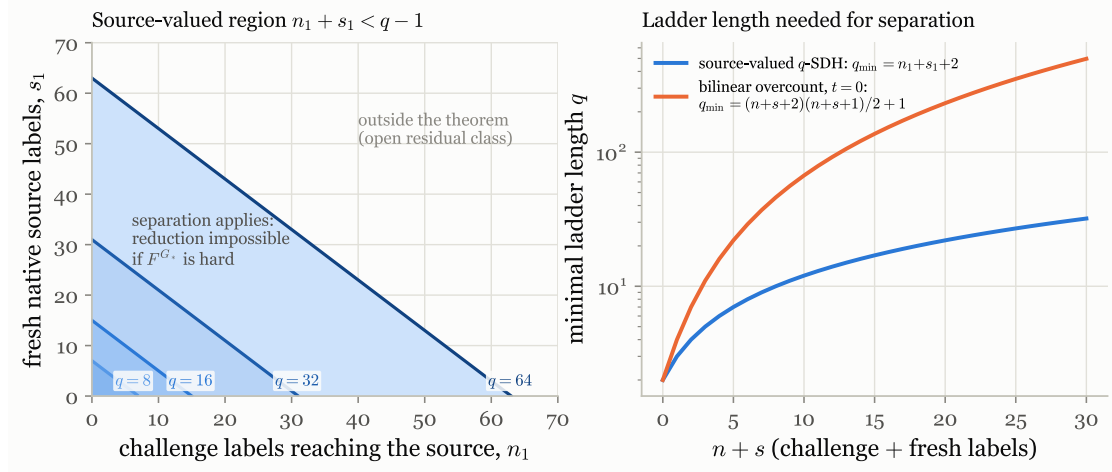


Figure 3: Parameter regions of the A006 thresholds; direct plots of the theorem’s inequalities, not data. Left: the source-valued separation region $n_1 + s_1 < q - 1$ for $q \in \{8, 16, 32, 64\}$ — deeper shading marks parameter pairs already separated at shorter ladders; above each line the residual class is open. Right: the minimal ladder length at which the separation bites, for the source-valued case ($q = n_1 + s_1 + 2$) versus the conservative bilinear overcount with $t = 0$ ($q = (n + s + 2)(n + s + 1)/2 + 1$): the safe bilinear bound demands quadratically longer ladders.

PROVED This closes a genuine part of the representation-dependent gap left by Lu–Zhandry: bounded native operations are now covered, and the remaining resource is precisely **source-label rank growing with q** — or an unrecorded conversion supplying it.

10 Dead ends and their exact failure points

Three attempted extensions failed. Each post-mortem narrows the open residual class, so we record them as results, not anecdotes.

10.1 Random relabeling cannot settle a fixed representation (A004)

Proposition 20 (the representation quantifier). **PROVED** A reduction guaranteed only for one family G_* has quantifier form

$$\exists G_* \exists R_* \forall A_* : \text{Break}_{Q^{G_*}}(A_*) \implies \text{Break}_{F^{G_*}}(R_*^{A_*}). \quad (7)$$

The random-relabeling simulation needs this implication for a freshly sampled random sparse implementation G_L , which the premise does not entail — and cannot be made to entail. Given any R_* working for a publicly specified G_* , modify it to check the canonical encoding of the public generator (or a public representation identifier), abort on mismatch, and otherwise run R_* : the modified machine behaves identically on G_* and can fail on every random relabeling. Abstract group isomorphism does not transfer the behavior of a machine that reads encodings.

Proposition 21 (the trace invariant fails syntactically). **PROVED** Independently, concrete encodings break the fixed-dimensional trace at the syntax level: native validation, decompression, or hash-to-curve code may produce a valid point without an observed group-operation query, so the simulator receives no algebraic explanation of that point in the span of the fixed-size challenge.

CITED The published structural discussion likewise leaves open derived group laws using auxiliary strings, equality branches, or concrete representations; the classification covers natural affine algebraic constructions only [5]. **HEURISTIC** Fresh representation-derived points do not visibly generate the correlated ladder $g^x, \dots, g^{x^q}$, so this trace failure is not positive evidence that a useful reduction exists; it only prevents the relabeling route from certifying impossibility. A006 is exactly the repair: charge the unexplained points instead of trying to relabel them away.

10.2 Efficient sparse encodings cannot replace random injections (A005)

Could the GR wrapper’s information-theoretic random injection be replaced by an **efficiently computable** sparse encoding, extending the separation to reductions promised over all efficient representations? No:

Theorem 22 (finite-seed counting obstruction). **PROVED** Let an efficient encoding family use an $m(\lambda)$ -bit seed, m polynomial, to choose injections from an r -element group into an N -element label space, $N \geq r$. A possibly inefficient adversary distinguishes the family from a uniform random injection with advantage at least $1 - 2^m / (N)_r$, where $(N)_r = N(N-1)\cdots(N-r+1)$ — an overwhelming bound, since $\log(N)_r \geq \log(r!) = \Omega(r \log r)$ while m is polynomial and r is exponential in λ .

Proof. The family’s support contains at most 2^m injective tables; the uniform distribution has support $(N)_r$. The distinguisher queries the encoding on all r inputs, reconstructs the full table, enumerates all 2^m seeds, and accepts exactly when the table belongs to the family: it accepts a family member with probability one and a random injection with probability at most $2^m / (N)_r$. $\square$

PROVED Neither proposed weakening survives. A t -wise independent encoding matches only a bounded-query view, and the fully-black-box adversary quantifier supplies no query bound — the distinguisher above uses $r > t$ queries. A PRP protects only computationally bounded distinguishers, so it buys a computational premise, not the needed information-theoretic wrapper. **CITED** This boundary accords with Zhandry’s generic-query convention, in which bit computation is unbounded and no efficient keyed permutation internal to the type-safe model is a secure PRP [16], and with the later sparse-GGM separations for admissibly encoded, dense, elliptic-curve, and bilinear settings [17].

PROVED A second, independent obstruction: breaking a static assumption in an artificial efficient encoding would refute hardness only **for that encoding**. If the assumption is asserted for a named curve family G_* , an attack on a different artificial family does not contradict its premise.

PROVED What does transfer is the query-bounded version: if the reduction class quantifies only over adversaries making at most t encoding or group queries, a t -wise or lazy-table simulation becomes plausible — a strictly narrower class that must state its query bound explicitly.

10.3 Structured-label density does not bound freshness (A007)

The structured generic-group model gives algorithms free access to a partial label operation $\star$ agreeing with the hidden group law, and charges only generic group-oracle queries. **CITED** For a prime-order- r group whose constrained-label fraction is δ , its hard labeling distribution bounds T -query discrete-log advantage by

$$\delta(3T + 2) + (3T + 1)^2/r + 1/r \quad (8)$$

[18]. Could density δ substitute for A006's freshness budget?

Proposition 23 (density is not a label count). **PROVED** The δT term bounds a **random-hybrid probability** — the chance a random challenge interaction touches the structured region — whereas freshness counts distinct labels **adaptively selected** along an actual execution. A publicly recognizable structured subset of size $\delta r \geq q$ lets a reduction deliberately address q distinct labels even when δ is negligible. The query parameter T cannot substitute either: $\star$ evaluations and all other label computation are free in the model's cost measure, so T does not bound the labels enumerated, parsed, or produced through $\star$ and then used as group elements.

PROVED Instantiating the model's hard labeling distribution instead repeats A004's quantifier error: the theorem is existential in a labeling distribution, and a reduction promised for one named labeling need not retain its guarantee there [18]. **PROVED** The valid conditional transfer is explicit and modest: if a separate premise bounds by u the total distinct valid labels first-used through raw code and free $\star$ evaluation, then A006 applies verbatim with $s = u$, and $n_1 + u < q - 1$ suffices for the source-valued separation. Density alone supplies no such u ; a future extension must bound native-label **rank** or exploit the algebraic relations revealed by $\star$ directly.

11 Scope classification and scheme evidence

Table 5 is the final classification: which reduction classes are ruled out, by what, and what survives.

Table 5: Final scope classification (NOTES §11). The strongest justified deliverable is the scoped impossibility of rows 1–4.

Reduction class	Result
CITED TS-BB / GR-BB, fully black-box in possibly inefficient adversaries	Ruled out at the published single-group ($n < q - 1$) and bilinear ($\binom{n+2}{2} < q$) thresholds [5].
CITED Algebraic reduction with explanations	The same tracing separation applies [5].
PROVED UR-FBB standard-oracle reduction, universal over representations	Implies GR-BB (Theorem 13, A003); ruled out at the same thresholds.
PROVED One fixed concrete representation, at most s_1 unexplained source labels	Direct trace separation at $n_1 + s_1 < q - 1$ (Theorem 16, A006), without random relabeling.
PROVED Representation class not closed under random sparse relabeling, native-label rank crossing A006's threshold	A004 blocks relabeling and A006 no longer applies; existence or impossibility open.
CITED Non-black-box use of the adversary's code	Outside the fully-black-box taxonomy and outside every theorem here [5], [15].

Scheme-level evidence reinforces the same picture from the outside: Table 6 summarizes the positive results and why none crosses the prime-order barrier.

Table 6: Positive results and scheme taxonomy (NOTES §6). Current BBS-family results retain q -dependent premises in both directions.

Setting	What is established	Why it does not close P2.2
CITED Déjà Q, 2014	Computational q -SDH and broad polynomial q -type classes reduce to constant-size subgroup hiding in composite-order pairings; the q -SDH proof uses $q + 2$ shadow copies with $\Theta(q)$ hybrid loss [13].	Not prime-order; asymmetric, computational-source only.
CITED Déjà Q follow-up, 2016	Broader classes, symmetric and asymmetric composite order, logarithmic tightness loss [14].	Still does not cross the prime-order barrier.
CITED Boneh–Boyen signatures	Standard-model proof under q -SDH with q the signing-query bound, roughly factor-two advantage loss [1].	A use of q -SDH, not a reduction of it.
CITED Dual-form exponent inversion	Modified Boneh–Boyen-like signatures and Gentry-IBE variants proved under SXDH [8].	Altered dual-form schemes — not the original assumption or scheme.
CITED BBS/BBS+ concrete security, 2025	After q signatures, attacks recover the key at $\Theta(q)$ -DL cost; scheme security implies $\Theta(q)$ -SDH [2].	A reverse, scheme-to- q -SDH comparison; reinforces usage-dependent security.
CITED BBS tightness, 2026	Tight q -SDH proof when each message is signed once; with repeats, no tight algebraic reduction to the considered q -SDH variants [3].	Tightens a q -type proof; neither removes q nor adds a static basis.

EMPIRICAL: literature searches, 2026-07-09 Primary-title, author, and ePrint-oriented searches found no later primary paper giving either an unrestricted prime-order reduction of q -SDH to a fixed-size assumption or an impossibility theorem covering every representation-specific black-box reduction.

12 Open problems

The residual gap is logged as P2.2/Q003 and has exactly two branches.

PROVED **Branch 1 — linear native rank.** No result here rules out a representation-dependent reduction that injects native-label rank linear in q : by the corollary of §9, a surviving reduction must

place at least $q - 1 - n_1$ unexplained labels in the q -SDH source in some execution, and nothing yet shows a concrete-operation class cannot do so. Closure requires either proving that every reduction in a precisely named class has source-label trace rank below $q - n_1 - 1$, or exhibiting a positive constant-size reduction whose native rank reaches the required dimension.

PROVED Branch 2 — code access. A reduction that reads the code of the q -SDH adversary is outside the fully-black-box taxonomy [15] and outside every theorem above. Closure requires an explicitly non-black-box construction or a meta-reduction whose scope includes code access.

PROVED Do not repeat. Random relabeling (A004), finite-seed substitution (A005), and density-to-count transfer (A007) each fail for the recorded, distinct reasons; the structural conclusion of the whole record is that the common obstruction is **dimensional** — a fixed-size prime-order challenge exposes a fixed-dimensional algebraic span, bilinearity raises the degree but leaves an $O(n^2)$ monomial space, and only composite-order hidden subgroups have so far manufactured the missing dimensions [5], [13]. The relevant resource for any future attack on Branch 1 is the rank of native labels actually injected into the adversary transcript, not their density, their efficiency, or their number in expectation.

13 Conclusion

P2.2 asked for a reduction or a separation. The answer supported by the record is a **layered separation with an honest boundary**. The catalogue and audited implication graph fix the objects (§3–5). Cheon’s attacks quantify the ladder’s intrinsic leak (§6). The direct constant-size embedding dies at x^2 for dimensional reasons (§7). The Lu–Zhandry theorem turns that dimension count into an impossibility for all fully black-box generic-representation reductions, and every representation-uniform standard-oracle reduction is inside that class (§8). Our main theorem extends the separation to a named concrete representation whenever the reduction’s unexplained native source labels are bounded, at the sharp source-valued threshold $n_1 + s_1 < q - 1$ and the safe bilinear overcount $\binom{n+s+2}{2} + t < q$ (§9). Three natural extensions fail at identified points, and their post-mortems narrow the residue to exactly two branches: native-label rank linear in q , and non-black-box use of the adversary’s code (§10–12). Within prime-order pairing groups, no route to basing q -SDH on a constant-size assumption survives except through one of those two doors — and no one has yet opened either.

Reproducibility

P2.2 is a theory problem: it contains no computational code and no data/ CSVs, and this paper reports none. The three figures are schematics generated by `papers/figures/P2.2/make.py` directly from the recorded content — the implication graph from the audited edge table (NOTES §2), the meta-reduction architecture from the proof of Theorem 16 (attempt A006), and the threshold regions from the stated inequalities $n_1 + s_1 < q - 1$ and $\binom{n+s+2}{2} + t < q$; none depicts measured data. Primary proof bodies checked during the research include four implication reductions, the Lu–Zhandry lemmas and main theorems, the Reingold–Trevisan–Vadhan fully-black-box definition, Zhandry’s labeling model, and the structured-GGM lower bound. Every mathematical claim above carries one of the epistemic tags **PROVED**, **CITED**, **HEURISTIC**, **EMPIRICAL: scope stated** exactly as in the research log; untagged sentences are exposition, not claims.

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